

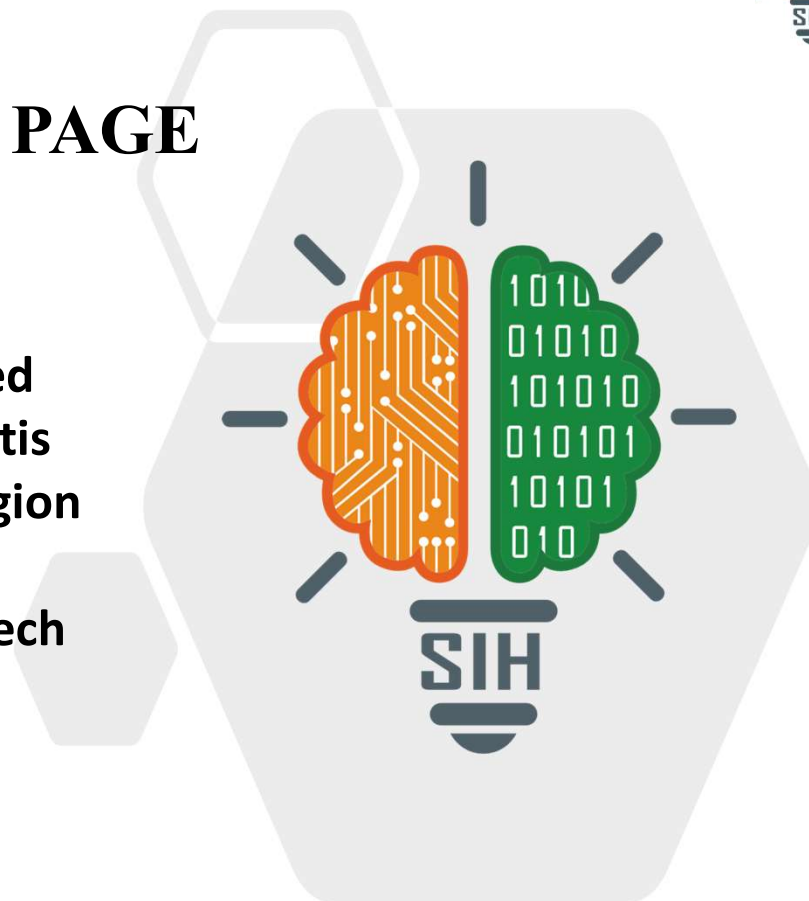
SMART INDIA HACKATHON 2026



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HACKATHON
2026

TITLE PAGE

- **Problem Statement ID –SIH26004**
- **Problem Statement Title- AI-Assisted Early Detection System for Osteoarthritis (OA) Risk Markers in North Eastern Region (NER).**
- **Theme- MedTech / BioTech / HealthTech**
- **PS Category- Hardware**
- **Team ID- 153685**
- **Team Name – Arogya-army**

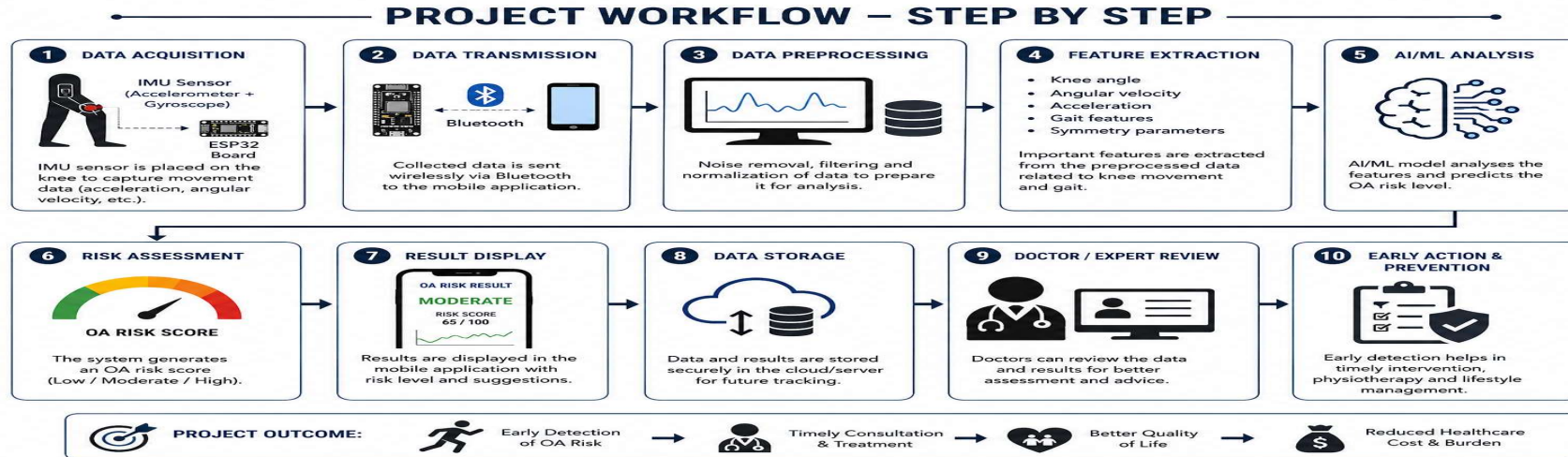


IDEA TITLE

❖ Proposed Solution

We propose a low-cost AI-assisted system for early detection of OA risk markers through:

- Wearable sensor-based gait and knee movement monitoring.
- AI/ML-based analysis of collected movement data.
- Mobile application for OA risk assessment and early alert.

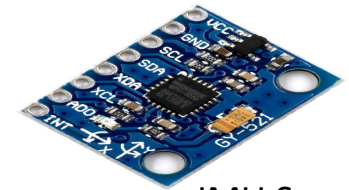


TECHNICAL APPROACH

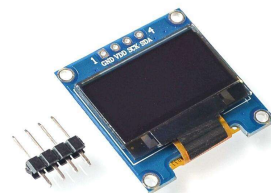


Here we are using following tech:-

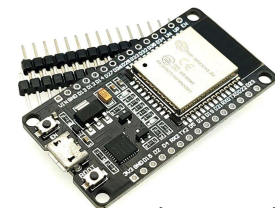
1. IMU Sensor (Accelerometer + Gyroscope)
2. ESP32 Development Board
3. AI/ML Model for OA Risk Prediction
4. 0.96" OLED Display



IMU Sensor

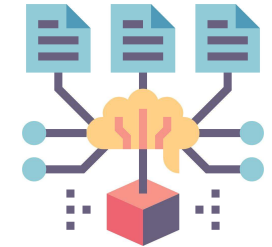


0.96" OLED



ESP32 Development board

FEASIBILITY AND VIABILITY



- **Feasibility of the Idea:**

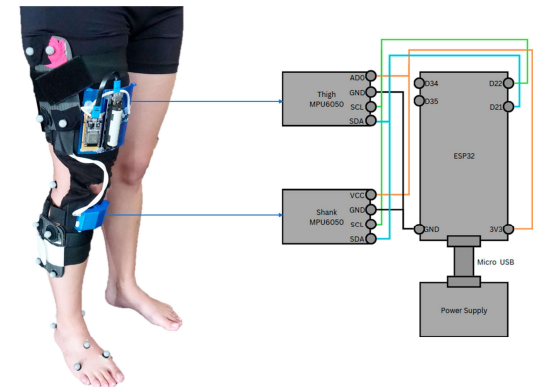
1. Low-cost and portable system using IMU and ESP32.
2. Easy to collect gait and knee movement data.
3. AI/ML can analyse movement patterns for early OA risk assessment.

- **Challenges are:**

1. Availability of reliable OA gait data.
2. Variation in sensor placement and user movement.
3. Accuracy and reliability of AI prediction.

- **Strategies for Overcoming Challenges:**

1. Use public datasets + collected real-world data.
2. Follow a fixed IMU placement and calibration method.
3. Train and validate the model with diverse user data.



IMPACT AND BENEFITS



- **Impact on Target Audience**

1. Enables early OA risk detection and promotes timely consultation.
2. Provides an affordable and accessible screening solution for NER.

- **Benefits**

1. **Social:** Improves OA awareness and preventive care.
2. **Economic:** Low-cost alternative for initial screening.
3. **Healthcare:** Supports doctors with movement-based risk data.

RESEARCH AND REFERENCES



1. Kobsar et al. (2020) – *Wearable Inertial Sensors for Gait Analysis in Adults with Osteoarthritis.* (<https://www.mdpi.com/1424-8220/20/24/7143>)

Supports the use of IMU sensors for OA gait analysis.

2. Fukushi et al. (2024) – *Gait Classification of Knee Osteoarthritis Patients Using Shoe-Embedded IMU sensors.*

(<https://pubmed.ncbi.nlm.nih.gov/38901396/>)

Demonstrates IMU + Machine Learning for OA classification.

3. TinyML-enabled wearable system (2025) – *Early Detection of Knee Osteoarthritis Using Ensemble Gait*

Classification. (<https://pubmed.ncbi.nlm.nih.gov/41353887/>)

Supports real-time wearable IMU + AI/ML-based OA detection.

4. WHO – Osteoarthritis Fact Sheet

Used for OA prevalence, risk factors and early management background.

(<https://www.who.int/news-room/fact-sheets/detail/osteoarthritis>)